

A full Besov-space characterization for generalized Cesàro-like operators

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2026-08-09

Status. Claimed full solution, likely valid, subject to human review. This packet answers the open problem in arXiv:2309.02717 for every $1 < p < \infty$ and every $\alpha > 0$. The proof replaces the Hilbert-space coefficient argument of the earlier partial packet by a dyadic analytic Besov decomposition.

1 Source question and answer

Let μ be a finite positive Borel measure on $[0, 1)$, let $\mu_n = \int_0^1 t^n d\mu(t)$, and put

$$c_r^{(\alpha)} = \frac{\Gamma(r + \alpha)}{\Gamma(\alpha)r!}.$$

For $f(z) = \sum_{k \geq 0} a_k z^k$, define

$$\mathcal{C}_{\mu, \alpha} f(z) = \sum_{n \geq 0} \mu_n \left(\sum_{k=0}^n c_{n-k}^{(\alpha)} a_k \right) z^n.$$

Tang [1] asks for a characterization of boundedness and compactness of $\mathcal{C}_{\mu, \alpha} : B_p \rightarrow B_p$, $p > 1$, where

$$\|f\|_{B_p} = |f(0)| + \left(\int_{\mathbb{D}} |f'(z)|^p (1 - |z|)^{p-2} dA(z) \right)^{1/p}.$$

For $N \geq 2$, define the tail quantity

$$\Phi_N(\mu; p, \alpha) = (\log(eN))^{p-1} \sum_{n=N}^{\infty} (n+1)^{p\alpha-1} \mu_n^p. \quad (1)$$

Theorem 1 (Full characterization). *Let $1 < p < \infty$ and $\alpha > 0$. Then:*

1. $\mathcal{C}_{\mu, \alpha}$ is bounded on B_p if and only if

$$\sup_{N \geq 2} \Phi_N(\mu; p, \alpha) < \infty.$$

2. $\mathcal{C}_{\mu, \alpha}$ is compact on B_p if and only if

$$\Phi_N(\mu; p, \alpha) \longrightarrow 0 \quad (N \rightarrow \infty).$$

At $p = 2$, this is the previously obtained Dirichlet criterion because $\log(eN) \asymp H_N$. At $p = 2, \alpha = 1$, it agrees with the later ordinary-kernel result of Bao–Guo–Sun–Wang [2].

Corollary 2 (Sharp boundary example). *For every $1 < p < \infty$ and $\alpha > 0$, the measure*

$$d\mu_\alpha(t) = \frac{(1-t)^{\alpha-1}}{\log(e/(1-t))} dt$$

induces a bounded, noncompact operator $\mathcal{C}_{\mu_\alpha, \alpha} : B_p \rightarrow B_p$.

2 Idea of the proof

The coefficient formula factors as

$$\mathcal{C}_{\mu, \alpha} = M_\mu K_\alpha, \quad K_\alpha f = (1-z)^{-\alpha} f, \quad M_\mu \left(\sum d_n z^n \right) = \sum \mu_n d_n z^n.$$

On a dyadic frequency block of size 2^j , the analytic Besov norm is $2^{j/p}$ times the Hardy norm. The j -th block of $(1-z)^{-\alpha}$ has Hardy norm of order $2^{j(\alpha-1/p)}$, while M_μ contributes $\mu_{2^{j-1}}$. Thus the output block sizes are dominated by

$$2^{j\alpha} \mu_{2^{j-1}} \sum_{\ell \leq j} x_\ell,$$

where $(x_\ell) \in \ell^p$ are the input Besov block sizes. This is exactly a weighted discrete Hardy operator. Its criterion is $\sup_m m^{p-1} \sum_{j \geq m} (2^{j\alpha} \mu_{2^j})^p < \infty$, which is equivalent to (1).

For necessity, normalized logarithmic polynomials have one unit of Besov mass on each of their first m dyadic blocks. Positivity makes all their output coefficients align on a short boundary arc, recovering the entire Hardy tail. These polynomials converge weakly to zero, so the same test also detects compactness.

3 Dyadic lemmas

For $j \geq 1$, set $I_j = \{2^{j-1}, \dots, 2^j - 1\}$, let $I_0 = \{0\}$, and write

$$\Delta_j f(z) = \sum_{n \in I_j} \widehat{f}(n) z^n.$$

Lemma 3 (Analytic Besov blocks). *For $1 < p < \infty$,*

$$\|f\|_{B_p}^p \asymp_p |\widehat{f}(0)|^p + \sum_{j \geq 1} 2^j \|\Delta_j f\|_{H^p}^p. \quad (2)$$

Moreover, the sharp interval projections Δ_j are uniformly bounded on H^p , and a polynomial with frequencies in an interval of length L satisfies

$$\|P\|_\infty \lesssim_p L^{1/p} \|P\|_{H^p}. \quad (3)$$

These are the standard dyadic description of analytic Besov spaces and the analytic Nikolskii inequality; they follow from the weighted Bergman norm of f' , uniform boundedness of interval projections for $1 < p < \infty$, and Bernstein's inequality. See, for example, the analytic Besov theory in [4].

Let $K_\alpha(z) = (1-z)^{-\alpha} = \sum_{r \geq 0} c_r^{(\alpha)} z^r$.

Lemma 4 (Kernel and moment blocks). *For every $\alpha > 0$, $1 < p < \infty$, and $j \geq 1$,*

$$\|\Delta_j K_\alpha\|_{H^p} \lesssim_{p,\alpha} 2^{j(\alpha-1/p)}, \quad (4)$$

$$\left\| \sum_{0 \leq r < 2^j} c_r^{(\alpha)} z^r \right\|_\infty \lesssim_\alpha 2^{j\alpha}. \quad (5)$$

If P has no frequencies below L , then

$$\|M_\mu P\|_{H^p} \leq \mu_L \|P\|_{H^p}. \quad (6)$$

Proof. Stirling's formula gives

$$c_r^{(\alpha)} \asymp_\alpha (r+1)^{\alpha-1}.$$

On I_j , Abel summation and

$$\left\| \sum_{k=0}^L z^k \right\|_{H^p} \asymp_p L^{1-1/p}$$

give (4). Summing the positive coefficients gives (5).

For (6), write $P(z) = z^L Q(z)$. Since radial dilation is contractive on H^p ,

$$\|P(t \cdot)\|_{H^p} = t^L \|Q(t \cdot)\|_{H^p} \leq t^L \|P\|_{H^p}.$$

Now use $M_\mu P(z) = \int_0^1 P(tz) d\mu(t)$ and Minkowski's inequality. □

We also use the classical discrete two-weight Hardy criterion [3].

Lemma 5 (Discrete Hardy criterion). *Let $1 < p < \infty$ and $a_j \geq 0$. The map*

$$(Tx)_j = a_j \sum_{\ell=0}^j x_\ell$$

is bounded on ℓ^p if and only if

$$\sup_{m \geq 1} m^{p-1} \sum_{j \geq m} a_j^p < \infty. \quad (7)$$

It is compact if and only if the products in (7) tend to zero.

The compactness statement follows by applying the same boundedness criterion after deleting the first J output coordinates. The resulting norm is controlled by

$$\sup_{m \geq 1} m^{p-1} \sum_{j \geq \max\{m, J\}} a_j^p,$$

which tends to zero exactly in the stated case.

4 Proof of the characterization

Sufficiency in Theorem 1. First take f to be a polynomial. Put

$$x_0 = |\hat{f}(0)|, \quad x_\ell = 2^{\ell/p} \|\Delta_\ell f\|_{H^p} \quad (\ell \geq 1).$$

By Lemma 3, $\|x\|_{\ell^p} \asymp \|f\|_{B_p}$.

Fix $j \geq 2$. If $\ell \leq j-2$, only the kernel frequencies comparable to 2^j can contribute to $\Delta_j(K_\alpha \Delta_\ell f)$. The boundedness of the output projection, (4), and (3) give

$$\|\Delta_j(K_\alpha \Delta_\ell f)\|_{H^p} \lesssim 2^{j(\alpha-1/p)} 2^{\ell/p} \|\Delta_\ell f\|_{H^p} = 2^{j(\alpha-1/p)} x_\ell.$$

For $\ell = j-1, j$, use (5); since $\ell \asymp j$, the same bound results. Inputs with $\ell > j$ cannot contribute because all kernel frequencies are nonnegative. Therefore

$$\|\Delta_j(K_\alpha f)\|_{H^p} \lesssim_{p,\alpha} 2^{j(\alpha-1/p)} \sum_{\ell=0}^j x_\ell. \quad (8)$$

The operators M_μ and Δ_j commute. Since the frequencies in I_j are at least 2^{j-1} , Lemma 4 yields

$$2^{j/p} \|\Delta_j(\mathcal{C}_{\mu,\alpha} f)\|_{H^p} \lesssim A_j \sum_{\ell=0}^j x_\ell, \quad A_j = 2^{j\alpha} \mu_{2^{j-1}}.$$

Lemma 5 now proves boundedness provided

$$\sup_{m \geq 1} m^{p-1} \text{sum}_{j \geq m} A_j^p < \infty. \quad (9)$$

Because (μ_n) is decreasing, dyadic grouping gives

$$\sum_{j \geq m} (2^{j\alpha} \mu_{2^j})^p \asymp_{p,\alpha} \sum_{n \geq 2^m} (n+1)^{p\alpha-1} \mu_n^p, \quad (10)$$

up to a fixed shift of m . Since $m \asymp \log(2^m)$, the hypothesis $\sup_N \Phi_N < \infty$ is equivalent to (9). Polynomials are dense in B_p , so the estimate extends $\mathcal{C}_{\mu,\alpha}$ to all of B_p . $\square$

Necessity in Theorem 1. Let $q = p/(p-1)$. For $m \geq 4$, define

$$F_m(z) = m^{-1/p} \sum_{k=1}^{2^{m-2}} \frac{z^k}{k}.$$

Abel summation and the Dirichlet-kernel H^p estimate show that each nonzero dyadic block satisfies

$$\|\Delta_\ell F_m\|_{H^p} \lesssim m^{-1/p} 2^{-\ell/p}.$$

Hence (2) gives $\sup_m \|F_m\|_{B_p} < \infty$.

Let $b_n = [z^n](\mathcal{C}_{\mu,\alpha} F_m)$. If $j \geq m$, $n \in I_j$, and $1 \leq k \leq 2^{m-2}$, then $n - k + 1 \asymp n + 1$. Positivity and the gamma-ratio estimate therefore give

$$b_n \gtrsim_\alpha m^{-1/p} \mu_n (n+1)^{\alpha-1} \sum_{k=1}^{2^{m-2}} \frac{1}{k} \gtrsim m^{1/q} \mu_n (n+1)^{\alpha-1}. \quad (11)$$

On an arc $|\theta| \leq c2^{-j}$, the numbers $e^{in\theta}$, $n \in I_j$, lie in a fixed acute sector. Thus their positive coefficients cannot cancel. Using monotonicity of μ_n and (11),

$$\|\Delta_j(\mathcal{C}_{\mu,\alpha} F_m)\|_{H^p}^p \gtrsim 2^{-j} m^{p-1} (2^{j\alpha} \mu_{2^j})^p.$$

After multiplying by the Besov block weight 2^j and summing over $j \geq m$, boundedness of $\mathcal{C}_{\mu,\alpha}$ forces

$$\sup_{m \geq 4} m^{p-1} \sum_{j \geq m} (2^{j\alpha} \mu_{2^j})^p < \infty.$$

By (10), this is precisely the condition $\sup_N \Phi_N < \infty$. $\square$

Compactness in Theorem 1. Suppose first that $\Phi_N \rightarrow 0$. Let Q_J delete the first $J - 1$ dyadic output blocks. The upper estimate and the tail form of Lemma 5 give

$$\|Q_J \mathcal{C}_{\mu, \alpha}\|_{B_p \rightarrow B_p}^p \lesssim \sup_{m \geq 1} m^{p-1} \sum_{j \geq \max\{m, J\}} A_j^p.$$

For $m < J$, this is at most $J^{p-1} \sum_{j \geq J} A_j^p$; for $m \geq J$, it is the tail supremum of the vanishing Hardy products. Hence $\|Q_J \mathcal{C}_{\mu, \alpha}\| \rightarrow 0$. The complementary output has finite dimension, so $\mathcal{C}_{\mu, \alpha}$ is compact.

Conversely, the bounded polynomials F_m tend to zero uniformly on compact subsets of $\mathbb{D}$, and hence weakly in the reflexive space B_p (use $B_p^* \cong B_q$ and density of polynomials). If $\mathcal{C}_{\mu, \alpha}$ is compact, then $\|\mathcal{C}_{\mu, \alpha} F_m\|_{B_p} \rightarrow 0$. The lower estimate from the necessity proof gives

$$m^{p-1} \sum_{j \geq m} (2^{j\alpha} \mu_{2^j})^p \rightarrow 0.$$

Equation (10) is equivalent to $\Phi_N \rightarrow 0$. □

5 Boundary example and checks

Proof of Corollary 2. With $u = 1 - t$, a standard Laplace estimate gives

$$\mu_{\alpha, n} = \int_0^1 \frac{(1-u)^n u^{\alpha-1}}{\log(e/u)} du \asymp_{\alpha} \frac{1}{(n+1)^{\alpha} \log(n+2)}.$$

Consequently,

$$\Phi_N(\mu_{\alpha}; p, \alpha) \asymp (\log N)^{p-1} \sum_{n \geq N} \frac{1}{(n+1) \log^p(n+2)} \asymp_p 1.$$

Theorem 1 gives boundedness and failure of compactness. □

As another consistency check, for Lebesgue measure and $\alpha = 1$, $\mu_n = (n+1)^{-1}$, so the series in (1) diverges. Thus the classical Cesàro operator is unbounded on every B_p , consistent with the previously known Dirichlet obstruction.

Computational check. The included script evaluates the moments of μ_{α} and truncated versions of Φ_N for several (p, α) . It confirms the predicted $n^{-\alpha}/\log n$ scale and nonzero boundary products. This check is a sanity test, not part of the proof.

Novelty check. On 2026-08-09 we searched the run indexes, the local arXiv source corpus, the source paper and its cited predecessor arXiv:2305.02717, and current web results using the exact operator formula, source title and arXiv id, the phrases “from B_p to B_p ”, and the proposed moment-tail weight. The search found the $p = 2, \alpha = 1$ theorem [2], range-space results, and later work on Dirichlet-type and Bloch-type spaces, but no self-map characterization for all p and α . Novelty confidence is bounded, not exhaustive.

Human-review recommendation. Verify Lemma 3 with the exact normalization used here and the two frequency-separated cases in (8). Those are the only non-scalar analytic inputs. The Hardy reduction, positive-arc lower bound, moment-tail equivalence and compactness argument are explicit.

References

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